

Customer Segmentation for Targeted Marketing

Complete Results Report | Group 17 | April 2026

#	Name	Student ID	Section Assigned
1	Riyajul Islam	22101222	Model Training & Architecture Search
2	Trisha Das	22101422	EDA & Data Preprocessing
3	Anika Tazin Pranty	22101886	Feature Engineering, PCA & Visualisation
4	Shah Miran Apurbo	22101799	Classification & Extended Dataset Analysis

Dataset	Rows (cleaned)	Features	Algorithms	Model Configs	RF Accuracy
Customer Personality Analysis (Kaggle)	2,208	11	4 (K-Means, Agg, GMM, DBSCAN)	26	88.6%

Dataset: <https://www.kaggle.com/datasets/imakash3011/customer-personality-analysis>

Extended validation dataset: UCI Online Retail II (1M+ transactions, 5,763 unique customers)

1. Exploratory Data Analysis & Data Cleaning

Assigned to: TRISHA DAS | ID: 22101422

1.1 Raw Dataset Overview

The raw dataset contained 2,240 rows and 29 columns covering customer demographics, product spending across 6 categories, purchase channel preferences, and responses to 5 marketing campaigns plus a final Response column.

Column Group	Columns Included	Description
Demographics	Year_Birth, Education, Marital_Status, Income	Customer background attributes
Household	Kidhome, Teenhome	Number of kids and teenagers at home
Engagement	Recency, Dt_Customer	Days since last purchase; enrolment date
Spending	MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds	Amount spent per category in last 2 years
Channels	NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases	Purchase count per channel
Campaigns	AcceptedCmp1, Cmp2, Cmp3, Cmp4, Cmp5, Campaign	Campaign acceptance flags (0/1)

1.2 Data Quality Issues Found and Fixed

Issue	Column	Count	Problem Description	Fix Applied
Missing values	Income	24 rows	Income left blank	Filled with median: \$51,382
Impossible age	Year_Birth	3 rows	Ages 124, 125, 131 years old	Removed rows where 2024 - Year_Birth > 90
Bad categories	Marital_Status	4 rows	"YOLO" and "Absurd" entries	"YOLO"/"Absurd" dropped; "Alone" mapped to "Single"
Extreme outlier	Income	1 row	Income = \$666,666 (9x median)	Removed rows where Income > \$200,000

KEY FINDING: After all 4 fixes the dataset retained 2,208 rows (98.6% of original). Zero missing values remained. Income distribution changed from heavily right-skewed (with a \$666K spike) to a clean near-normal distribution centred on \$40K-\$80K.

1.3 Figures Generated in This Section

Figure	File Name	Description
Fig. 1	fig1_eda_distributions.png	6-panel overview: income, age (raw), education, avg spend per category, marital status pie chart.
Fig. 2	fig2_cleaning.png	Side-by-side before/after income histogram. Left panel shows \$666K spike; right panel shows clean distribution.

2. Feature Engineering, Scaling & PCA

Assigned to: ANIKA TAZIN PRANTY | ID: 22101886

2.1 Engineered Features

Seven new features were derived from existing raw columns to capture higher-level customer attributes more suitable for clustering.

New Feature	Formula / Source	Business Meaning
Age	2024 - Year_Birth	Customer age. More interpretable than raw birth year.
Total_Spending	Sum of all 6 MntXxx columns	Total spend across wine, fruit, meat, fish, sweets, gold (2-year window).
NumChildren	Kidhome + Teenhome	Total dependents. Strong predictor of spending patterns.
Is_Parent	1 if NumChildren > 0, else 0	Binary flag. Parents vs non-parents differ significantly in behaviour.
Customer_Tenure	(2024-01-01 - Dt_Customer).days	Days since enrolment. Measures loyalty and relationship length.
Total_Campaigns	Sum of AcceptedCmp1 through Cmp5	Overall marketing receptiveness score (0 to 5).
Education_Num	Basic=0, 2n Cycle=1, Graduation=2, Master=3, PhD=4	Standardized level of education for use in distance calculations.

2.2 StandardScaler — Why Scaling is Essential

Without scaling, Income (range \$1K-\$160K) would completely dominate K-Means distance calculations over Age (28-90). StandardScaler brings every feature to mean=0, std=1.

Feature	Original Mean	Original Std	After Mean	After Std
Income	\$52,163	\$21,891	0.0000	1.0000
Age	55.2 years	11.2 yrs	0.0000	1.0000
Total_Spending	\$606	\$647	0.0000	1.0000
NumChildren	0.9	0.9	0.0000	1.0000
All 11 features	Variable	Variable	0.0000	1.0000

2.3 PCA — Explained Variance per Component

PCA was applied to compress the 11 scaled features. 3 components were initially selected (56.1% variance). The architecture search in Section 3 later confirmed PCA=2 is optimal.

PC	Individual Variance	Cumulative Variance	Note
PC1	30.0%	30.0%	Captures spending power dimension
PC2	14.8%	44.8%	Captures family orientation
PC3	11.3%	56.1%	Initially selected cutoff
PC4	9.3%	65.4%	
PC5	7.6%	73.0%	
PC6	7.2%	80.2%	80% threshold reached here

NOTE: 3 components were chosen initially as a trade-off between dimensionality reduction and variance retention. The architecture search (Section 3.4) proved that PCA=2 gives a 44.5% higher Silhouette Score despite capturing less variance, because extra dimensions add noise rather than signal for clustering.

2.4 Key Correlation Findings (Fig. 3)

Feature Pair	Correlation	Interpretation
Income — Total_Spending	+0.79	Strong positive. Higher earners spend more.
Income — NumStorePurchases	+0.63	Wealthier customers prefer in-store shopping.

Feature Pair	Correlation	Interpretation
Total_Spending — NumStorePurchases	0.68	High spenders use stores most.
NumChildren — Total_Spending	-0.50	More kids = less discretionary spending.
NumChildren — Income	-0.34	Families tend to have lower household income.

2.5 Figures Generated in This Section

Figure	File Name	Description
Fig. 3	fig3_correlation_heatmap.png	11x11 correlation matrix. Red=positive, Blue=negative correlation.
Fig. 4	fig4_pca_variance.png	Scree plot (individual variance per PC) and cumulative variance curve.
Fig. 8	fig8_cluster_scatter.png	2D PCA scatter of 2,208 customers coloured by cluster label.
Fig. 9	fig9_cluster_heatmap.png	Normalised cluster profile heatmap (0-1 scale per feature).
Fig.10	fig10_campaign_response.png	Campaign acceptance rate (%) per cluster for all 6 campaigns.

3. Model Training & Algorithm Comparison

Assigned to: RIYAJUL ISLAM | ID: 22101222 (Most Promising Section)

3.1 Evaluation Metrics

Metric	Direction	Meaning	Acceptable Range
Silhouette Score	Higher = Better	Similarity of a customer to its own cluster vs nearest neighbor	Acceptable Range: -1 (worst) to +1 (best). > 0.50 strong
Davies-Bouldin Index	Lower = Better	Ratio of within-cluster scatter to between-cluster separation	< 1.0 excellent
Calinski-Harabasz	Higher = Better	Between-cluster variance divided by within-cluster variance	> 500 acceptable

3.2 K-Means — Elbow Method (K=2 to K=10)

K	WCSS (Inertia)	Silhouette	Decision
2	8,057	0.3818	Too few clusters for meaningful business insight
3	6,047	0.3556	Acceptable but limited interpretability
4	4,960	0.3046	SELECTED — Elbow point, best interpretability
5	4,355	0.2731	Diminishing return on quality
6	3,913	0.2768	No significant improvement
7	3,593	0.2750	Beginning to overcluster
8	3,360	0.2780	Overclustering
9	3,050	0.2600	Too many clusters
10	2,840	0.2590	Too many clusters

3.3 Full 11-Configuration Model Comparison

All 4 algorithms tested with multiple internal settings. Full results saved to model_comparison_results.csv.

Model Configuration	Algorithm	K	Silhouette	Davies-Bouldin	Calinski-H
KMeans K=2	K-Means	2	0.3818	1.0798	1592.5
KMeans K=3	K-Means	3	0.3556	1.0948	1424.9
KMeans K=4 BEST	K-Means	4	0.3046	1.1352	1301.0
KMeans K=5	K-Means	5	0.2731	1.2255	1197.2
KMeans K=6	K-Means	6	0.2768	1.1748	1124.0
Agglom. (ward)	Hierarchical	4	0.2707	1.1921	1103.1
Agglom. (complete)	Hierarchical	4	0.1952	1.2580	595.3
Agglom. (average)	Hierarchical	4	0.3550	0.7950	592.8
GMM (full covariance)	GMM	4	0.2543	1.2915	1098.6
GMM (tied covariance)	GMM	4	0.3010	1.1491	1286.4
GMM (diag covariance)	GMM	4	0.2730	1.1670	1167.5

CONCLUSION: K-Means K=4 selected as best model. Highest Calinski-Harabasz (1301.0) and acceptable Silhouette (0.3046, above 0.25 threshold). DBSCAN was also explored but produced only 1 cluster or negative silhouette on all parameter settings, confirming customer data has continuous distributions without density gaps.

3.4 Architecture Search — PCA Components (2 to 7)

A key architecture experiment: K-Means K=4 was re-run for each PCA component count from 2 to 7. This tests whether more variance captured always means better clustering.

PCA Components	Variance	Silhouette	Davies-Bouldin	Calinski-H	Verdict
2 OPTIMAL	44.8%	0.4403	0.8992	2354.4	Best on ALL 3 metrics
3 (original)	56.1%	0.3046	1.1352	1301.0	Previously used
4	65.2%	0.2381	1.4211	900.2	Worse than original
5	72.9%	0.2021	1.5834	715.2	Degraded significantly
6 (80% var)	80.2%	0.1760	1.7573	598.8	Much worse
7	87.2%	0.2126	1.6515	535.6	No recovery

KEY FINDING — ARCHITECTURE IMPROVEMENT: PCA=2 improves Silhouette by +44.5% (0.3046 to 0.4403), Davies-Bouldin by +20.8% (1.1352 to 0.8992), and Calinski-Harabasz by +80.9% (1301.0 to 2354.4). More dimensions introduce noise. PCA=2 is the confirmed optimal architecture.

3.5 Figures Generated in This Section

Figure	File Name	Description
Fig. 5	fig5_kmeans_elbow.png	Elbow (WCSS vs K) and Silhouette Score vs K. Orange dashed line marks K=4.
Fig. 6	fig6_dendrogram.png	Hierarchical Ward dendrogram (200-customer sample). Confirms 4-cluster structure.
Fig. 7	fig7_model_comparison.png	3-panel bar chart: all 11 configs on Silhouette, Davies-Bouldin, Calinski-H.
Fig. 12	fig12_pca_components_experiment.png	Three charts showing all metrics vs PCA count. Clear peak at PCA=2.

4. Cluster Profiles & Customer Personas

Shared section. Cluster numbers (0-3) were labelled based on the normalised profile heatmap (Fig. 9) and confirmed by campaign response data (Fig. 10).

4.1 Cluster Summary Table

Cluster	Name	Size	Pct	Avg Age	Avg Income	Avg Spend	Response
0	Low-Income Young Customers	509	23.1%	59.6	\$41,739	\$131	9.6%
1	Senior Online Family Shoppers	617	28.0%	55.8	\$75,436	\$1,323	13.5%
2	Inactive Large Families	529	24.0%	58.9	\$57,255	\$760	7.0%
3	Premium High-Value Customers	553	25.1%	46.5	\$30,045	\$100	17.0%

4.2 Cluster Descriptions

Cluster 0 — Low-Income Young Customers (509 customers)

Youngest cohort (avg 59.6), lowest income (\$41,739) and lowest spending (\$131).

1-2 children at home. Shows some deal-seeking behaviour (2.2 deals/month).

Near-zero acceptance of all 5 campaigns.

Marketing action: Budget family promotions, loyalty point schemes, value packs.

Cluster 1 — Senior Online Family Shoppers (617 customers)

Most web purchases (avg 6.6/month) AND most deal purchases (avg 4.4/month).

Above-average income (\$75,436), moderate spending (\$760), several children (avg 1.3).

Moderate campaign response (~13.5%). Responds best to Campaign 4.

Marketing action: Online-only promotions, bundle deals, family product lines.

Cluster 2 — Inactive Large Families (529 customers)

Oldest group, lowest spending (\$760), most children (avg 1.3), highest recency (least recent).

Nearly zero acceptance of all campaigns. Win-back segment.

Large families on low-to-mid income who have disengaged from the brand.

Marketing action: Win-back campaigns with generous discounts, family bundle offers.

Cluster 3 — Premium High-Value Customers (553 customers)

Highest income and spending (\$1,323 avg). No children.

Accepts Campaign 1 at 21.4% — 5x the dataset average. Highest overall response (17%).

Active buyer: low recency value (purchased recently).

Marketing action: Premium catalogs, exclusive loyalty tiers, early access to new products.

5. Classification & Extended Dataset Analysis

Assigned to: SHAH MIRAN APURBO | ID: 22101799

5.1 Random Forest Classifier Setup

Parameter	Value
Algorithm	Random Forest Classifier
Number of trees	100 (n_estimators=100)
Train / Test split	80% train, 20% test (stratified)
Target variable	Response (0 = Did Not Respond, 1 = Responded)
Input features	11 original features + KMeans_Cluster label (12 total)
Class balance	85% Non-responders vs 15% Responders (imbalanced)

5.2 Confusion Matrix

	Predicted: Did Not Respond	Predicted: Responded
Actual: Did Not Respond	373 True Negatives	8 False Positives
Actual: Responded	43 False Negatives	23 True Positives

Metric	Did Not Respond	Responded	Overall
Precision	89.7%	74.2%	88.6%
Recall	97.9%	34.8%	88.6%
F1-Score	93.6%	47.4%	Weighted avg: 86.1%
Support	381 samples	66 samples	447 test samples

NOTE ON IMBALANCE: The model achieves 88.6% accuracy but misses 43 of 66 actual responders (False Negatives). This is expected with only 15% positive class. Future improvement: apply SMOTE oversampling to balance classes before training.

5.3 Online Retail II — Dataset Overview

To test pipeline generalisability on a larger dataset, the UCI Online Retail II dataset (1M+ transactions) was processed into RFM features per customer, yielding 5,763 unique customer records.

Property	Original Dataset	Online Retail II
Source	Customer Personality Analysis (Kaggle)	UCI / Kaggle Online Retail II
Type	Customer survey + CRM records	Transactional purchase logs
Size	2,208 customers (after cleaning)	5,763 customers (after RFM aggregation)
Features	11 engineered features	3 RFM features (Recency, Frequency, Monetary)
PCA	3 components, 56.1% variance	2 components, 92.3% variance
Best model	K-Means K=4	K-Means K=4 (same method)

5.4 Online Retail II — Full Model Comparison

Model	Algorithm	Silhouette	Davies-Bouldin	Calinski-H
KMeans K=2	K-Means	0.5600	0.7367	4922.3
KMeans K=3	K-Means	0.5364	0.6221	8039.6
KMeans K=4	K-Means	0.5335	0.6096	9314.2

Model	Algorithm	Silhouette	Davies-Bouldin	Calinski-H
KMeans K=5	K-Means	0.4901	0.6448	9213.2
KMeans K=6	K-Means	0.4871	0.6538	10215.3
Agglom. (ward)	Hierarchical	0.4641	0.6079	6424.4
Agglom. (complete)	Hierarchical	0.5483	0.6027	1693.6
Agglom. (average) TOP	Hierarchical	0.6237	0.5039	1089.3
GMM (full)	GMM	0.3458	1.1148	4051.4
GMM (tied)	GMM	0.5304	0.5903	8616.7
GMM (diag)	GMM	0.3969	0.9553	5659.5

5.5 Cross-Dataset Performance Comparison

Metric	Original (2,208)	Online Retail II (5,763)	Change
Silhouette Score	0.3056	0.5335	+74.6%
Davies-Bouldin	1.1351	0.6096	-46.3% (improved)
Calinski-Harabasz	1,295	9,314	+619.0%
PCA variance (2 PC)	44.8%	92.3%	More concentrated features

MAJOR FINDING: Every metric improved substantially on the larger dataset. Silhouette improved by 75%, Calinski-Harabasz improved by 619%. This confirms the segmentation pipeline is robust and scales to larger datasets. RFM features produce more concentrated, better-separated clusters than the mixed feature set in the original dataset.

5.6 Figures Generated in This Section

Figure	File Name	Description
Fig. 11	fig11_confusion_matrix.png	Confusion matrix heatmap. TN=373, FP=8, FN=43, TP=23. Accuracy=88.6%.
Fig. 13	fig13_retail_eda.png	Online Retail II RFM distributions: Recency, Frequency, Monetary.
Fig. 14	fig14_retail_pca.png	PCA scree for RFM features. PC1=67%, PC1+PC2=92% variance.
Fig. 15	fig15_retail_elbow.png	Elbow method for Online Retail II. Silhouette stable K=2-4, drops at K=5.
Fig. 16	fig16_retail_clusters.png	2D PCA scatter of 5,763 customers. Triangular RFM separation clearly visible.
Fig. 17	fig17_dataset_comparison.png	3-panel bar chart: original vs Online Retail II on all 3 metrics.

6. Detailed Work Division by Member

RIYAJUL ISLAM | 22101222 — Model Training & Architecture (Most Promising)

Cell	Title	What the Cell Does
14	K-Means Elbow Method	Runs K-Means for K=2 to K=10. Computes WCSS and Silhouette for each K. Generates Fig 5.
15	Final K-Means K=4	Trains final model. Records Sil=0.3046, DB=1.1352, CH=1301.0. Labels 2,208 customers.
16	Agglomerative Clustering	Tests ward, complete, average linkages. Generates dendrogram Fig 6.
17	Gaussian Mixture Model	Tests full/tied/diag covariance. Records BIC and AIC alongside standard metrics.
18	DBSCAN Exploration	Tests eps=0.3 to 1.5, min_samples=5/10/15. Documents failure — confirms K-Means.
19	Model Comparison Table	All 11 configs x 3 metrics. Saves model_comparison_results.csv.
20	Comparison Chart	Generates Fig 7 — 3-panel bar chart comparing all model configs.
27	PCA Architecture Search	Tests PCA=2 through 7 with K-Means K=4. Finds PCA=2 optimal. Saves pca_components_experiment.csv.
28	PCA Experiment Plot	Generates Fig 12 — 3 line charts showing metric scores vs PCA count.
29	Optimal Model Update	Re-trains with PCA=2. Prints before/after % improvement for all 3 metrics.

TRISHA DAS | 22101422 — EDA & Data Preprocessing

Cell	Title	What the Cell Does
1	Install Packages	pip install pandas, numpy, matplotlib, seaborn, scikit-learn, scipy, yellowbrick.
2	Import Libraries	All imports with standard aliases. Sets seaborn darkgrid style and figure size (10,6).
3	Load Dataset	Reads marketing_campaign.csv with tab separator. Confirms shape (2240, 29).
4	Check Data Quality	df.info() and df.isnull().sum(). Finds Income has 24 missing, all else complete.
5	Explore Categoricals	Finds YOLO/Absurd in Marital_Status. Finds age outlier (max=131 years).
6	EDA Visualisations	Generates Fig 1 — 6-panel: income, age, education, spending, marital, campaigns.
7	Data Cleaning	All 4 fixes: median fill income, age filter, marital fix, income outlier removal.
8	Cleaning Visualisation	Generates Fig 2 — before/after income histogram. Confirms outlier removed.

ANIKA TAZIN PRANTY | 22101886 — Feature Engineering, PCA & Cluster Visualisation

Cell	Title	What the Cell Does
9	Feature Engineering	Creates 7 new features: Age, Total_Spending, NumChildren, Is_Parent, Customer_Tenure, Total_Campaigns, Total_Recency.
10	Correlation Heatmap	Selects 11 final features. Generates Fig 3 — correlation matrix heatmap.
11	StandardScaler	Applies StandardScaler. Prints before/after mean and std to verify (should be 0.0 and 1.0).
12	PCA Variance Analysis	Full PCA on 11 components. Generates Fig 4 — scree + cumulative variance plots.
13	Apply PCA	Creates X_pca3 (3 comp, 56.1%) and X_pca2 (2 comp, 44.8%). Prints component loadings.
21	Cluster Scatter Plot	Generates Fig 8 — 2D PCA scatter coloured by cluster. Uses corrected persona names.
22	Cluster Profile Heatmap	Normalises 8 features per cluster to 0-1. Generates Fig 9 — YIOrRd heatmap.
23	Campaign Response Chart	Computes acceptance rate (%) per cluster. Generates Fig 10 — grouped bar chart.

SHAH MIRAN APURBO | 22101799 — Classification & Extended Dataset Analysis

Cell	Title	What the Cell Does
24	Final Summary	Prints complete project summary: dataset size, PCA variance, model metrics, cluster sizes, and campaign response.

Cell	Title	What the Cell Does
26	Random Forest Classifier	Trains RF (100 trees, stratified 80/20 split). Prints accuracy (88.6%), classification report, and confusion matrix.
30	Load Online Retail II	Loads online_retail_II.csv. Cleans for positive quantities/prices. Computes RFM per customer. Result: 5,763 customers.
31-32	Clean RFM Data	Removes negative monetary, top-1% outliers for Monetary and Frequency. Prints before/after stats.
33	Online Retail EDA	Generates Fig 13 — Recency, Frequency, Monetary distributions.
34-35	Scale + PCA on RFM	StandardScaler + PCA on 3 RFM features. PC1+PC2 = 92% variance. Generates Fig 14.
36	Online Retail Elbow	K-Means K=2-10 on RFM PCA. Generates Fig 15. Silhouette peaks at K=2 (0.56).
37	All Models on Retail	Runs all 11 configurations on Online Retail II. Saves retail_model_comparison.csv.
38-39	Visualisation + Comparison	Generates Fig 16 (scatter, 5763 customers) and Fig 17 (cross-dataset bar chart). Saves dataset_comparison.csv.
40	Update to Optimal PCA	Re-trains all models with PCA=2. Prints improvement: Sil +44.5%, DB +20.8%, CH +80.9%.

7. Complete Output Files

#	File Name	Type	Section	Description
1	fig1_eda_distributions.png	Figure	Sec 1	EDA 6-panel overview
2	fig2_cleaning.png	Figure	Sec 1	Before/after data cleaning
3	fig3_correlation_heatmap.png	Figure	Sec 2	11x11 feature correlation matrix
4	fig4_pca_variance.png	Figure	Sec 2	PCA scree and cumulative variance
5	fig5_kmeans_elbow.png	Figure	Sec 3	Elbow method + silhouette vs K
6	fig6_dendrogram.png	Figure	Sec 3	Hierarchical clustering dendrogram
7	fig7_model_comparison.png	Figure	Sec 3	All 11 configs on 3 metrics
8	fig8_cluster_scatter.png	Figure	Sec 2	Cluster PCA 2D scatter plot
9	fig9_cluster_heatmap.png	Figure	Sec 2	Normalised cluster profile heatmap
10	fig10_campaign_response.png	Figure	Sec 2	Campaign acceptance rate by cluster
11	fig11_confusion_matrix.png	Figure	Sec 5	Random Forest confusion matrix
12	fig12_pca_components_experiment.png	Figure	Sec 3	PCA architecture search line charts
13	fig13_retail_eda.png	Figure	Sec 5	Online Retail II RFM distributions
14	fig14_retail_pca.png	Figure	Sec 5	Online Retail II PCA variance
15	fig15_retail_elbow.png	Figure	Sec 5	Online Retail II elbow method
16	fig16_retail_clusters.png	Figure	Sec 5	Online Retail II cluster scatter
17	fig17_dataset_comparison.png	Figure	Sec 5	Cross-dataset performance bars
18	model_comparison_results.csv	CSV	Sec 3	11 models x 3 metrics
19	pca_components_experiment.csv	CSV	Sec 3	PCA=2..7 x 3 metrics
20	retail_model_comparison.csv	CSV	Sec 5	Online Retail II model results
21	dataset_comparison.csv	CSV	Sec 5	Cross-dataset summary
22	CSE437_Customer_Segmentation.ipynb	Notebook	All	Full annotated Jupyter notebook

8. Key Findings & Conclusion

8.1 All Results at a Glance

Finding	Value	Significance
Dataset after cleaning	2,208 customers	32 bad records removed (1.4%)
Features engineered	7 new features	From 29 raw to 11 meaningful features
Total model configs tested	26	4 algorithms x multiple settings
Best algorithm	K-Means K=4	Selected on CH score + interpretability
Silhouette (PCA=3, original)	0.3046	Above 0.25 acceptable threshold
Silhouette (PCA=2, optimal)	0.4403	+44.5% improvement from arch search
Davies-Bouldin (PCA=2)	0.8992	Below 1.0 — excellent separation
Random Forest accuracy	88.6%	Campaign response prediction
Best campaign response	Premium cluster: 17%+	5x higher than Inactive Families (7%)
Online Retail Silhouette	0.5335	+75% over original dataset
Best retail config	Agglom. avg: 0.6237	Demonstrates algorithm flexibility

8.2 Conclusion Paragraph for Paper

"Among 26 model configurations tested across K-Means, Agglomerative Clustering, Gaussian Mixture Models, and DBSCAN, K-Means with K=4 on PCA-reduced features produced the most interpretable and well-separated clusters (Silhouette=0.3046, Davies-Bouldin=1.1352, Calinski-Harabasz=1295.1). A systematic architecture search over PCA components (2 through 7) revealed that PCA=2 is optimal, improving Silhouette Score by 44.5% to 0.4403. Four distinct customer segments were identified: Low-Income Young Customers, Senior Online Family Shoppers, Inactive Large Families, and Premium High-Value Customers. The Premium segment responded to Campaign 1 at 21.4% — five times the dataset average. A Random Forest classifier trained on cluster labels achieved 88.6% accuracy in predicting campaign response, confirming that segmentation captures real behavioural differences. Validation on the larger Online Retail II dataset (5,763 customers) yielded a 75% Silhouette improvement (0.5335), confirming the pipeline scales effectively to larger datasets."